**Module 5 Assignment: Variant Interpretation of CYP2C19**

BIOT 6900
Computational Lab 5
Ajay Kanaparthy
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## 6.1 Thresholds and calibration

I worked on CYP2C19, a gene that affects how the body processes drugs such as clopidogrel. I fetched 80 variants. Both CYP2C19*2 and CYP2C19*17 were found, so the main quality check passed. CADD scores were available for 68 variants, REVEL for 54, and gnomAD frequencies for 55. Not every variant had all the annotations.

I chose 0.50 as my maximum allele-frequency threshold. CYP2C19 is a pharmacogene, so a common variant can still affect drug response. A rare-disease cutoff is not suitable here. My 0.50 filter removed 2 variants and kept 78. Neither removed variant was labelled pathogenic/likely pathogenic (P/LP) or drug response in my data. The 0.001 cutoff would have removed 19 variants, including rs4244285 (CYP2C19*2), which had a frequency of 0.167953 in my dataset. This is a known no-function allele, so removing it just because it is common would be a mistake (Scott et al., 2012).

I selected CADD because it covered 85.0% of my original variants, compared with 67.5% each for REVEL and AlphaMissense. However, there were only 5 benign/likely benign (B/LB) variants and 0 P/LP variants available for calibration. The CADD band (0, 10] had all 5 benign variants and gave an apparent likelihood ratio (LR+) of 0.00, labelled "strong, benign" by the code. The other four bands had no labelled variants, so their likelihood ratios were undefined (NaN). I cannot trust these values because there is no pathogenic comparison group and every band has fewer than 10 labelled variants. I did not use them as evidence to change any VUS classification. All 33 VUS stayed VUS.

## 6.2 Coverage gaps and the regulatory genome

The coverage table had 54 missense variants. AlphaMissense scored all 54, and REVEL scored 53. Both missense predictors had no scores for the intergenic, intronic, splice-donor, stop-gained, stop-lost, synonymous, and upstream-gene classes. There were also 8 variants with missing consequence annotations. A blank score means the tool did not assess the variant. It does not mean the variant is benign.

One example from my output is rs7902257. It was listed as an upstream_gene_variant, with a gnomAD frequency of 0.022133. CADD, REVEL, and AlphaMissense were all blank for this row. To investigate it, I would test whether the change affects CYP2C19 expression in liver-derived cells. For example, an allele-specific expression assay or a promoter-reporter assay could

give functional evidence. I would also need evidence linking that change to enzyme activity or drug response before making a clinical interpretation.

I skipped the AlphaGenome Atlas because I did not configure an API key. This fallback is allowed in the assignment, and I do not have Atlas scores or top_feature results to report. My notebook identified 25 non-missense variants after filtering. Atlas could have helped investigate possible regulatory effects for rs7902257 and possible splicing effects for the intronic variant rs12769205. If MERGED_SPLICING drove a score, I would investigate RNA splicing. If MAX_ABS_DNASE, CHIP_TF, or RNA_SEQ drove it, I would look at regulatory activity and relevant tissues. A conservation-driven feature such as CACTUS_241_WAY would not by itself explain the mechanism. SpliceAI can predict splice changes, but REVEL and AlphaMissense cannot assess noncoding expression changes, and neither gives the same tissue-linked regulatory picture as Atlas.

**6.3 One discordant variant: rs4244285**

My discordance analysis flagged 21 unique variants. The counts were 3 for CADD versus REVEL, 6 for REVEL versus AlphaMissense, 0 for high-scoring VUS, 4 for common P/LP-or-other variants, and 10 with no tool score. Some variants can meet more than one rule, so the counts should not be added together.

I chose rs4244285 (CYP2C19*2). It triggered flag_common_but_plp because its frequency was above 0.01 and my notebook labelled it "other" (drug response). The flag name does not mean ClinVar called it P/LP in my data. The consequence was synonymous_variant, the CADD score was 5.686, REVEL and AlphaMissense returned no score, and the gnomAD frequency was 0.167953. Atlas was skipped, so there was no Atlas result.

The explanation I support is a splicing defect. The CYP2C19*2 change is c.681G>A (p.Pro227=). It does not directly change the amino acid, but published evidence shows that it creates an abnormal splice site and can produce a truncated, nonfunctional CYP2C19 protein (Scott et al., 2012). That explains why the low CADD score and blank missense scores do not match the important drug-response effect. The important biology is the RNA change, not an amino-acid substitution. I would give more weight to the established splicing evidence than to the low CADD score alone. This is about enzyme function and drug response, not a diagnosis made from a predictor score.

**6.4 What this analysis cannot determine**

First, this analysis cannot tell me a person's zygosity or phase. I only examined individual variants. Without knowing both CYP2C19 alleles, I cannot assign a person's complete diplotype or metabolizer status. For example, CYP2C19*1/*2 and *2/*2 do not have the same predicted phenotype (Lee et al., 2022).

Second, it cannot tell me exactly how one patient will respond to a medicine. Other drugs, enzyme inhibitors, health conditions, and other factors can change the observed response. This is called phenoconversion. A sequence score by itself cannot settle those clinical effects (Lee et al., 2022).

The single experiment I would choose is RT-PCR analysis of CYP2C19 RNA from liver-derived cells carrying rs4244285, followed by sequencing the PCR products. I would compare the variant cells with a reference-genotype control. If I see an abnormal splice product with the variant, it would support my explanation. If repeated testing shows normal splicing, the same as the control, I would reconsider the mechanism and check for another linked change. This experiment would test the splice mechanism, but it would not by itself prove a patient's drug response.

## References

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